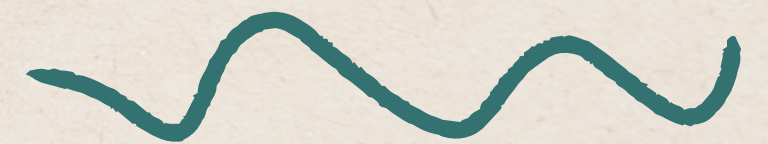




SkyBook



Project Name

SkyBook – Flight Booking & Airline Management System

Project Description

SkyBook is a web-based application developed using ASP.NET Core MVC that allows customers to search for flights, select seats, make reservations, and manage their bookings. The system also provides an administration panel for managing airports, aircrafts, flights, customers, and bookings.

Project Objectives

Simplify flight reservation process.

Provide an easy way to manage flights.

Allow customers to book seats online.

Give administrators full control over airline operations.

Functional Requirements

FR-01: User Registration

Description

The system shall allow new users to create an account.

Inputs

Full Name

Email Address

Password

Confirm Password

Business Rules

Email address must be unique.

Password must contain at least 6 characters.

Password and Confirm Password must match.

Output

A new customer account is successfully created.

Importance

This feature enables users to access the flight booking services.

FR-02: User Login

Description

The system shall allow registered users to log into the system.

Inputs

Email Address

Password

Business Rules

Only registered users can log in.

Invalid credentials must be rejected.

Output

User is authenticated and redirected to the appropriate page.

Importance

Authentication is required to access customer-specific features.

FR-03: Flight Search

Description

The system shall allow customers to search for available flights.

Inputs

Departure Airport

Arrival Airport

Travel Date

Business Rules

Departure and arrival airports cannot be the same.

Only available flights should be displayed.

Output

A list of matching flights is displayed.

Importance

This is the primary feature of the system.

FR-04: View Flight Details

Description

The system shall display detailed information about a selected flight.

Information Displayed

Flight Number

Departure Airport

Arrival Airport

Departure Time

Arrival Time

Aircraft

Available Seats

Ticket Price

Flight Status

Output

A detailed flight information page.

Importance

Helps customers make informed booking decisions.

FR-05: Seat Selection

Description

The system shall allow customers to choose a seat before booking.

Business Rules

Occupied seats cannot be selected.

Seat availability must be displayed in real time.

Output

The selected seat is assigned to the booking process.

Importance

Ensures accurate seat allocation.

FR-06: Flight Booking

Description

The system shall allow customers to reserve a seat on a selected flight.

Inputs

Flight ID

Seat Number

Business Rules

The selected flight must be available.

The selected seat must not be occupied.

Process

Validate flight availability.

Validate seat availability.

Create booking record.

Generate booking reference.

Update seat status.

Output

Booking confirmation with a unique booking reference.

Importance

Core functionality of the system.

FR-07: View My Bookings

Description

The system shall allow customers to view all their bookings.

Information Displayed

Booking Reference

Flight Number

Travel Date

Seat Number

Booking Status

Output

A list of customer bookings.

Importance

Allows customers to track their reservations.

FR-08: View Booking Details

Description

The system shall display complete details of a selected booking.

Information Displayed

Booking Reference

Passenger Information

Flight Information

Seat Information

Booking Status

Output

Detailed booking information.

Importance

Provides complete reservation details.

FR-09: Cancel Booking

Description

The system shall allow customers to cancel a booking.

Business Rules

A booking cannot be cancelled after the flight departure time.
Cancelled seats become available again.

Output

Booking status is updated to "Cancelled".

Importance

Allows customers to manage their reservations.

Admin Functional Requirements

FR-10: Airport Management

Actor

Admin

Description

The system shall allow administrators to manage airport records.

Functions

Add Airport

Edit Airport

Delete Airport

View Airports

Importance

Airports are required for flight creation.

FR-11: Aircraft Management

Actor

Admin

Description

The system shall allow administrators to manage aircraft information.

Functions

Add Aircraft

Edit Aircraft

Delete Aircraft

View Aircraft

Importance

Aircrafts are assigned to flights.

FR-12: Flight Management

Actor

Admin

Description

The system shall allow administrators to manage flights.

Functions

Create Flight

Edit Flight

Delete Flight

View Flights

Business Rules

Departure and arrival airports must be different.

Arrival time must be later than departure time.

Aircraft must be assigned to the flight.

Importance

This is the most critical administrative feature.

FR-13: Booking Management

Actor

Admin

Description

The system shall allow administrators to monitor and manage bookings.

Functions

View All Bookings

Search Bookings

Cancel Bookings

View Passenger Information

Importance

Provides operational control over reservations.

FR-14: Customer Management

Actor

Admin

Description

The system shall allow administrators to manage customer accounts.

Functions

View Customers

Disable Customer Accounts

View Customer Activity

Importance

Ensures proper management of system users.

OhFR-15: Dashboard

Actor

Admin

Description

The system shall provide a dashboard containing key statistics.

Information Displayed

Total Customers

Total Flights

Total Airports

Total Aircrafts

Total Bookings

Importance

Provides a quick overview of system performance.

System Requirements (Business Rules)

SR-01

The system shall generate a unique booking reference for every reservation.

SR-02

The system shall prevent double booking of the same seat.

SR-03

The system shall automatically update seat availability after booking.

SR-04

The system shall restore seat availability after booking cancellation.

SR-05

The system shall prevent creation of flights with past dates.

SR-06

The system shall prevent selection of the same airport as both departure and arrival airport.

SR-07

Only users with the Admin role may access the administration panel.

SR-08

Customers may only view and manage their own bookings.

Non Functional Requirements

Security

Password Hashing

Authentication

Authorization

Performance

Fast search results.

Efficient database queries.

Usability

Simple UI.

Responsive Design.

System Actors

Customer

Can do:

Register

Login

Search Flights

Book Flight

Cancel Booking

Admin

Can do:

Manage Airports

Manage Aircrafts

Manage Flights

Manage Bookings

View Dashboard

Use Case: Book Flight

Actor

Customer

Steps

Login

Search Flight

Select Flight

Select Seat

Confirm Booking

Result

Booking is successfully created.

.Database Design

Airport

Purpose

Stores airport information.

Fields

Id

Name

Code

City

Country

Aircraft

Purpose

Stores aircraft information.

Fields

Id

Name

Capacity

Flight	Seat	Booking
Purpose	Purpose	Purpose
Stores flight schedules.	Stores aircraft seats.	Stores reservation information.
Fields	Fields	Fields
Id	Id	Id
FlightNumber	SeatNumber	UserId
DepartureAirportId	Class	FlightId
ArrivalAirportId	AircraftId	SeatId
AircraftId		BookingDate
DepartureTime		Status
ArrivalTime		BookingReference
Price		
Status		

Relationships

Airport → Flight

One-To-Many

.Airport can be used in many flights

Aircraft → Flight

One-To-Many

.Aircraft can perform many flights

Aircraft → Seat

One-To-Many

.Aircraft contains many seats

User → Booking

One-To-Many

.User can create multiple bookings

Flight → Booking

One-To-Many

.Flight can have many bookings

Thank you